PROJECT NAME: The "Safe-Night" Transit

Problem Overview:

In many university campuses, transportation services stop operating after late evening hours (typically after 10 PM). However, a significant number of students continue to stay in libraries, labs, and study areas until late at night (up to 2 AM). This creates a serious mobility and safety challenge, as students are often left without reliable transportation options to return to their hostels or residences.

Existing System Issues:

Currently, universities rely on a **manual security escort system**, where students must call a central office to request a ride. This system has several limitations:

- Long waiting times (30–40 minutes)
- Limited availability of vehicles (often only one van)
- No real-time tracking or ETA updates
- No ride pooling, leading to inefficient vehicle usage
- High dependency on manual coordination
- Safety concerns due to delays or lack of availability

These issues result in students either waiting for extended periods or choosing to walk alone at night, which increases safety risks.

Core Problem Statement:

The core problem is to design a **safe, efficient, and scalable transportation system** that can provide on-demand mobility to students during late-night hours while optimizing limited resources and ensuring user safety.

Key Requirements Identified:

From the problem analysis, the system must:

- Provide **on-demand ride booking** within campus
- Support **ride pooling** to maximize vehicle efficiency
- Offer **real-time updates** such as ETA
- Ensure **student safety** through verification and emergency features
- Minimize **waiting time** and improve service reliability
- Operate within **campus boundaries** using limited vehicles

Constraints:

- Must use **open-source and cost-effective technologies**
- Should focus on **system design and logic**, not heavy UI

- Needs to be **scalable and practical** for real campus deployment

Understanding Summary:

The problem is not just about transportation, but about building a **smart, secure, and efficient mobility system** that addresses both logistical challenges and student safety concerns. The solution must balance **resource optimization (few vehicles)** with **high demand (many students)** while maintaining reliability and responsiveness.